Lecture 9

Search, and why it cannot be faster

Objectives. *Lecture 8 found factors, with no classical lower bound to compare. This lecture asks:* how much faster than classical can unstructured search be, and could it be faster still? *Grover's algorithm finds one marked item among N with about $\frac{\pi}{4}\sqrt{N}$ queries (Theorem 9.1); a classical algorithm needs at least $2N/3 - 1$ (Proposition 9.2). The main result is a lower bound: every quantum algorithm needs at least $(\sqrt{2N/3} - 1)/2$ (Theorem 9.5). With a bound proved on each side comes the course's second answer.*

By the end of this lecture you will be able to:

- Prove that Grover's iteration rotates a plane by 2θ, $\sin\theta = 1/\sqrt{N}$, and compute its success after ℓ iterations, a count at the first peak and the cost of overshooting.
- Prove that a classical algorithm with bounded error needs at least $2N/3 - 1$ queries.
- Prove by the hybrid argument that every quantum algorithm with bounded error and any workspace needs at least $(\sqrt{2N/3} - 1)/2$ queries.
- State the status of both bounds, and why "trying all answers at once" misdescribes the algorithm.

9.1 Reflections and the setup

Definition 5.1 decided a property of f; *search* finds something. Let $N = 2^n$, $n \geq 1$. For $w \in \{0,1\}^n$ let $f_w(x) = 1$ if $x = w$, and 0 otherwise: f_w *marks* the item w. The promise is that f is one of the N functions f_w, and the task is to return w. The cost counts queries, and an algorithm succeeds *with bounded error* if it returns w with probability at least $\frac{2}{3}$ for every w.

With the target in $|-\rangle$, the oracle acts on the query register as

$$O_w = \mathbb{1} - 2\,|w\rangle\langle w|\,,$$

by kickback (2.7). It reverses $|w\rangle$ and fixes every vector orthogonal to it: a *reflection*. The second reflection is about the uniform state $|u\rangle = H^{\otimes n}|0\cdots0\rangle = N^{-1/2}\sum_x |x\rangle$ (Proposition 5.3):

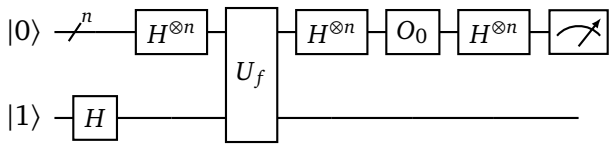
$$R_u = 2\,|u\rangle\langle u| - \mathbb{1}$$

fixes $|u\rangle$ and reverses every vector orthogonal to it. It needs no query. As $H^{\otimes n}$ is its own inverse and maps $|0\cdots0\rangle$ to $|u\rangle$,

$$\begin{aligned} R_u &= H^{\otimes n}\big(2\,|0\cdots0\rangle\langle0\cdots0| - \mathbb{1}\big)H^{\otimes n} \\ &= -H^{\otimes n}O_0H^{\otimes n}, \end{aligned}$$

where O_0 is O_w for $w = 0\cdots0$: the kickback of the oracle of the known function $f_{0\ldots0}$, which Theorem 1.9 builds with clean ancillas.

Grover's iteration is $G = R_u O_w$: one query, then a reflection that uses none. The algorithm prepares $|u\rangle$ and the target $|-\rangle$, applies ℓ iterations, and reads the query register:



The four boxes from U_f on form one iteration, drawn once and applied ℓ times. Each applies $-G$, a global phase that changes no probability (Theorem 3.7).

9.2 The rotation, one marked item

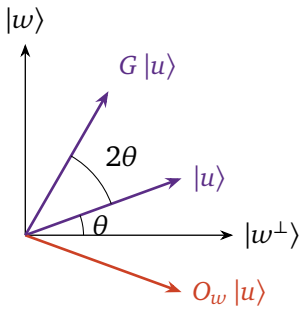
Two reflections make a rotation. Let

$$\left|w^{\perp}\right\rangle = (N-1)^{-1/2} \sum_{x \neq w} |x\rangle ,$$

a unit vector orthogonal to $|w\rangle$, and let θ be the angle with $\sin\theta = 1/\sqrt{N}$ and $0 < \theta \leq \pi/4$ ($\theta = \pi/4$ only for $N = 2$). In the plane of $|w^{\perp}\rangle$ and $|w\rangle$ write

$$v(\alpha) = \cos\alpha \left|w^{\perp}\right\rangle + \sin\alpha\, |w\rangle$$

for real α. As $\cos\theta = \sqrt{(N-1)/N}$, $|u\rangle = v(\theta)$. Nielsen and Chuang's θ (their (6.12), p. 252) is twice ours. The diagram shows the reflections in the plane of $|w^{\perp}\rangle$ and $|w\rangle$:



The first reflection sends $|u\rangle$ below the horizontal axis. Reflecting about $|u\rangle$ then gives $G|u\rangle$, at angle 3θ.

Theorem 9.1 (Grover's algorithm, one marked item). *Let $N = 2^n \geq 2$ and let w be marked.*

(i) For every $\ell \geq 0$, $G^{\ell}|u\rangle = v((2\ell+1)\theta)$, so ℓ iterations, read, return w with probability $\sin^2((2\ell+1)\theta)$.

(ii) With $\ell^* = \lfloor \pi/(4\theta) \rfloor \leq \frac{\pi}{4}\sqrt{N}$ iterations they return w with probability at least $1 - 1/N$.

Proof of (i). First, O_w reverses $|w\rangle$ and fixes $|w^{\perp}\rangle$, so $O_w v(\alpha) = v(-\alpha)$. Next, $R_u v(\beta) = 2\langle u|v(\beta)\rangle\,|u\rangle - v(\beta)$, with $\langle u|v(\beta)\rangle = \cos(\beta-\theta)$, and the product formulas for $2\cos(\beta-\theta)\cos\theta$ and $2\cos(\beta-\theta)\sin\theta$ give

$$2\cos(\beta-\theta)\, v(\theta) = v(\beta) + v(2\theta-\beta).$$

Hence $R_u v(\beta) = v(2\theta-\beta)$, and $Gv(\alpha) = R_u v(-\alpha) = v(\alpha+2\theta)$: G rotates the plane by 2θ, and $G^{\ell}|u\rangle = v((2\ell+1)\theta)$. The circuit leaves $(-1)^{\ell} G^{\ell}|u\rangle \otimes |-\rangle$, and reading the query register gives w with probability $\left|\langle w|\, G^{\ell}|u\rangle\right|^2 = \sin^2((2\ell+1)\theta)$. ∎

The first iterations turn the state towards $|w\rangle$, by 2θ each. The query alone finds nothing: $O_w|u\rangle$ differs from $|u\rangle$ only in the sign of one amplitude, and is read as w with probability $1/N$, as $|u\rangle$ is. The reflection R_u turns that sign into amplitude, by interference.

9.3 Success, the first peak, overshoot

Proof of Theorem 9.1(ii). As $\pi/(4\theta) - 1 < \ell^* \le \pi/(4\theta)$, the angle $(2\ell^*+1)\theta$ is within θ of $\pi/2$. On $[0,\pi]$, $\sin^2$ falls with the distance from $\pi/2$, and $\pi/2+\theta < \pi$, so the success is at least $\sin^2(\pi/2-\theta) = \cos^2\theta = 1 - 1/N$. Finally $\theta \ge \sin\theta = 1/\sqrt{N}$, so $\ell^* \le \frac{\pi}{4}\sqrt{N}$. ∎

For $N \ge 4$ the success is at least $\frac{3}{4}$. For $N = 2$, $\theta = \pi/4$ and every ℓ gives $\frac{1}{2}$, no better than a guess; one classical query decides between two items.

Past ℓ^*, the first peak, the state turns on, past $|w\rangle$, and the success first falls; it is near 0 when $(2\ell+1)\theta$ is near π, about $\ell = 2\ell^*$. For $N = 2^{20}$, $\ell^* = 804$ iterations return w with probability above $1 - 3\cdot10^{-7}$, and 1608 with probability below 10^{-10}. A classical search keeps what it has found; the rotation does not, so ℓ is chosen from N before the run. With several marked items the angle changes; here there is just one.

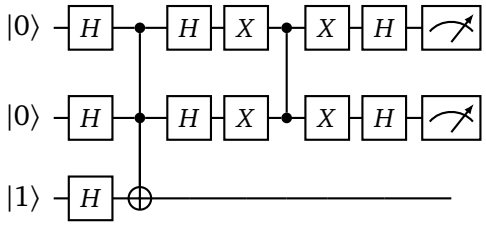
9.4 $N = 4$

For $N = 4$, $\sin\theta = \frac{1}{2}$, so $\theta = \pi/6$ and $\ell^* = 1$: one iteration turns $|u\rangle$ to the angle $3\theta = \pi/2$, onto $|w\rangle$. For $w = 11$, $O_w|u\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ has the overlap $\frac{1}{2}$ with $|u\rangle$, so

$$R_u O_w |u\rangle = |u\rangle - O_w|u\rangle = |11\rangle\,.$$

One query finds w with certainty. A second overshoots to $5\theta = 5\pi/6$: $G|11\rangle = \frac{1}{2}(-|00\rangle - |01\rangle - |10\rangle + |11\rangle)$ is read as 11 with probability $\frac{1}{4}$, as $|u\rangle$ is. Nielsen and Chuang's Box 6.1 (p. 256) has $\theta = \pi/3$, twice ours, and the count $\lceil \pi\sqrt{N}/4 \rceil$ of their algorithm (p. 254), taken literally, is 2 here: it overshoots.

The oracle of f_{11} is the Toffoli, and O_0 on two qubits is $(X\otimes X)\,\mathrm{CZ}\,(X\otimes X)$. The whole run, read on the top two wires, returns 11 with certainty:



A classical algorithm with one query succeeds, for a uniform w, with probability at most $\frac{1}{2}$ (Proposition 9.2).

9.5 The hybrid argument

Could another algorithm do better? First the classical competitor, by an argument that the quantum proof repeats.

Proposition 9.2 (Classical search)**.** *A classical query algorithm with at most k queries returns w, for a uniform w, with probability at most $(k+1)/N$. If it returns w with probability at least $\frac{2}{3}$ for every w, then $k \ge 2N/3 - 1$.*

Proof. For $k \geq N-1$ there is nothing to prove. Otherwise fix its random bits, and follow the run while every answer is 0. As $k < N$, while it has at most k such answers some string w is unasked; the run on f_w is this one so far, and makes at most k queries, so this run stops after at most k and returns a string. For every w neither asked nor returned, f_w answers 0 to each string asked, so the run on f_w is this run and does not return w. So with these bits it succeeds for at most $k+1$ of the N values of w; averaged over the bits and a uniform w, with probability at most $(k+1)/N$. If every w succeeds with probability at least $\frac{2}{3}$, so does a uniform w. ∎

The quantum proof also compares each run with the run that sees no marked item, on the constant function 0, whose oracle is $\mathbb{1}$; as one query may ask every string, it follows the weight on w, not the strings asked.

Algorithms. The register has $n+1+a$ qubits: the query register, the target, and a workspace of any size a. A *k-query algorithm* applies gates $V_0, V_1, \ldots, V_k$, independent of f, to $|0\cdots0\rangle$, with U_f on the first $n+1$ wires between each two; it then reads the whole register and returns a function g of the reading. On f_w its states are

$$\psi_0^w = V_0 \,|0\cdots0\rangle\,,$$
$$\psi_t^w = V_t U_{f_w} \psi_{t-1}^w \quad (t = 1, \ldots, k),$$

and ψ_t is the same with every query replaced by $\mathbb{1}$. Let $\Pi_w = |w\rangle\langle w| \otimes \mathbb{1}$, so that $\|\Pi_w \psi\|^2$ is the probability that a reading of the query register gives w.

Lemma 9.3 (Readings at the end)**.** *If a quantum query algorithm makes at most k queries in every run on every f of its promise, then some k-query algorithm returns each string with the same probability as it, on every f of the promise.*

Proof. Prepare the registers and record outcomes. Join every register a run may use at the start, in $|0\cdots0\rangle$, and give it its state by a gate when the run would join it. A projective measurement $\Pi_1, \ldots, \Pi_K$ (Definition 3.4) becomes the map

$$\psi \otimes |0\cdots0\rangle \mapsto \textstyle\sum_i \Pi_i \psi \otimes |i\rangle$$

onto fresh record wires, followed by a reading of them. The map keeps inner products, as the Π_i are orthogonal and sum to $\mathbb{1}$, so it extends to a gate, by completing orthonormal bases; the reading gives i with probability $\|\Pi_i \psi\|^2$ and, when this is positive, leaves the Lüders state times $|i\rangle$.

Defer the record readings. Each step chosen by earlier outcomes becomes that step controlled by their records, which no later step touches otherwise, so Theorem 4.5 moves every reading to the end.

Align and pad the queries. For $j = 1, \ldots, k$, each run's controlled steps go up to its j-th query, or to its end if it has none, controlled swaps bring its query wires to fixed wires, one U_f acts there, and the swaps are undone; a run with fewer than j queries has an idle $|0\cdots0\rangle\,|+\rangle$ there, which U_f fixes. The remaining steps come after the k-th. A run that would make a $(k+1)$-th query has probability 0 on every f of the promise, and is stopped there. ∎

Lemma 9.4 (The hybrid argument)**.** *For every k-query algorithm and every w,*

$$\left\|\psi_k^w - \psi_k\right\| \leq 2 \textstyle\sum_{t=0}^{k-1} \|\Pi_w \psi_t\|\,,$$

and

$$\textstyle\sum_w \left\|\psi_k^w - \psi_k\right\|^2 \leq 4k^2.$$

Proof. U_{f_w} applies X to the target exactly when the query register holds w, so $U_{f_w} - \mathbb{1} = |w\rangle\langle w| \otimes (X - \mathbb{1}) \otimes \mathbb{1}$, which is $X - \mathbb{1}$ on the target after Π_w. $X - \mathbb{1}$ has the eigenvalues 0 and -2, so its norm is 2, and by §6.1 a matrix on some wires has no larger norm on the whole register. So

$$\left\|(U_{f_w} - \mathbb{1})\psi\right\| \le 2\left\|\Pi_w\psi\right\|$$

for every ψ. Now

$$\begin{aligned}\psi_t^w - \psi_t &= V_t U_{f_w}(\psi_{t-1}^w - \psi_{t-1}) \\ &\quad + V_t(U_{f_w} - \mathbb{1})\psi_{t-1},\end{aligned}$$

and gates keep lengths, so $\delta_t = \left\|\psi_t^w - \psi_t\right\|$ has

$$\delta_t \le \delta_{t-1} + 2\left\|\Pi_w\psi_{t-1}\right\|.$$

As $\delta_0 = 0$, summing over t gives the first claim. By Cauchy–Schwarz over t, its right side squared is at most $4k\sum_t \|\Pi_w\psi_t\|^2$. The Π_w project onto orthogonal subspaces that together span the register, so $\sum_w \|\Pi_w\psi_t\|^2 = \|\psi_t\|^2 = 1$, and summing over w gives $4k \cdot k$. ∎

The lemma is a form of Bennett, Bernstein, Brassard and Vazirani's Theorem 3.3 (pp. 7–8). Its proof is the telescoping of Theorem 6.2 applied to one state; applied to the gates, it would give only $2k$, as $\left\|U_{f_w} - \mathbb{1}\right\| = 2$. The gain is in the state: ψ_t does not depend on w, so it cannot put much weight on every w. In Grover's algorithm the run on 0 is $\psi_t = \pm|u\rangle|-\rangle$, with the weight $1/N$ on each w, and the distance grows by at most $2/\sqrt{N}$ per query.

9.6 Success forces distance

Theorem 9.5 (Search needs $\sqrt{N}$ queries)**.** *Let $N = 2^n$. If a quantum query algorithm, with any workspace, returns w with probability at least $\frac{2}{3}$ for every w, then some run makes k queries with*

$$k \ge \left(\sqrt{2N/3} - 1\right)/2.$$

Proof. By Lemma 9.3 it has the output distribution of a k-query algorithm. Let E_w project onto the readings z with $g(z) = w$; the E_w project onto orthogonal subspaces. The run on f_w succeeds with probability $\left\|E_w\psi_k^w\right\|^2$, and a projector does not lengthen a vector, so by the triangle inequality

$$\sqrt{2/3} \le \left\|E_w\psi_k\right\| + \left\|\psi_k^w - \psi_k\right\|.$$

Sum over w. Cauchy–Schwarz bounds the sum involving the output projectors:

$$\begin{aligned}\textstyle\sum_w \left\|E_w\psi_k\right\| &\le \sqrt{N}\left(\textstyle\sum_w \left\|E_w\psi_k\right\|^2\right)^{1/2} \\ &\le \sqrt{N}.\end{aligned}$$

For the state distances, Cauchy–Schwarz and Lemma 9.4 give

$$\textstyle\sum_w \left\|\psi_k^w - \psi_k\right\| \le \sqrt{N} \cdot 2k.$$

Hence

$$N\sqrt{2/3} \le \sqrt{N}(1 + 2k).$$

Rearranging gives the stated query lower bound. ∎

The run on 0 does not know w, so it returns a given w with probability $1/N$ on average; to return w with probability $\frac{2}{3}$, the run on f_w must move far from it, for most w, and on average over w that distance grows by at most $2/\sqrt{N}$ per query. For $N = 4$ the lower bound is 1 query, which Grover's one iteration meets. For $N = 2^{20}$ it is 418 queries, against Grover's 804 and the classical 699 050. Grover's algorithm is optimal up to a constant. Nielsen and Chuang (pp. 269–270) prove a lower bound for success $\frac{1}{2}$ without a workspace, by comparing with basis vectors; its constant is not ours.

9.7 Status and the second answer

Both bounds are *proved*, unconditionally, in the query model: quantum search needs of the order of $\sqrt{N}$ queries, classical search of the order of N (asking $N - 1$ strings suffices). Like Simon's (Lecture 5), the comparison is relative to an oracle; unlike it, the gap is quadratic.

It says nothing about problems given by a circuit, where an algorithm may read the circuit and the query lower bound does not apply. To find the one input accepted by a known circuit, Grover's algorithm makes about $\frac{\pi}{4}\sqrt{N}$ queries, each built from the circuit (Theorem 1.9), against up to $N - 1$ runs for trying every input: a quadratic speed-up over exhaustive search, not over every classical algorithm. Whether NP, the decision problems whose yes-instances have a certificate that a classical algorithm checks in polynomial time, lies in BQP is open, and the query lower bound decides nothing about it (Bennett, Bernstein, Brassard and Vazirani, p. 3).

The course asks what makes the difference visible. Its second answer: *the difference is interference arranged by structure, counted against a stated classical model, and bounded below.* Simon's algorithm and phase estimation arrange interference by a group; Grover's by two reflections. The lower bound refutes "trying all answers at once": one query on $|u\rangle$ writes every value of f into a phase, yet a reading after it returns w with probability $1/N$, as a guess does. The hybrid argument bounds what interference can do: on average over w, the distance from the run that sees no marked item grows by at most $2/\sqrt{N}$ per query.

Summary.

- *Grover:* the iteration rotates a plane by 2θ, $\sin\theta = 1/\sqrt{N}$; ℓ iterations succeed with probability $\sin^2((2\ell+1)\theta)$, at least $1 - 1/N$ at the first peak, $\ell = \lfloor \pi/(4\theta) \rfloor$; past it, overshoot (Theorem 9.1).
- *Classical:* bounded error needs at least $2N/3 - 1$ queries (Proposition 9.2).
- *The hybrid argument:* $\sum_w \left\| \psi_k^w - \psi_k \right\|^2 \leq 4k^2$ (Lemma 9.4).
- *The main result:* bounded error needs at least $(\sqrt{2N/3} - 1)/2$ queries, with any workspace (Theorem 9.5): unconditional, in the query model.
- *Next:* Lecture 10, errors and the first codes.

References

Nielsen and Chuang, pp. 248–256 and 269–271. De Wolf, pp. 53–56 and 89. Watrous (2025), pp. 240–250. L. K. Grover, *A fast quantum mechanical algorithm for database search*, arXiv:quant-ph/9605043 (1996). C. H. Bennett, E. Bernstein, G. Brassard and U. Vazirani, *Strengths and weaknesses of quantum computing*, arXiv:quant-ph/9701001 (1997), pp. 3 and 7–8.